

Exploratory Data Analysis & Model Building

K.L. Khavriya varshini
192472381

1. Data Collection

- Collect data from different sources
- Structured & unstructured data
- Ensure data quality



2. Data Preprocessing

- Handle missing values
- Remove duplicates
- Detect outliers
- Clean and transform data
- Scale features



3. Exploratory Data Analysis

- Summary statistics
- Data visualization
- Correlation analysis
- Find patterns
- Detect anomalies



4. Feature Engineering

- Select important features
- Extract new features
- Reduce dimensionality
- Create meaningful representations



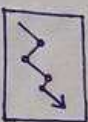
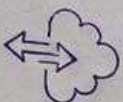
5. Model Building

- Train - test split
- Cross validation
- Choose algorithm
- Train the model
- Tune parameters



Deployment & Monitoring

- Deploy the model
- Monitor performance
- Update the model
- Feedback loop



Model Evaluation

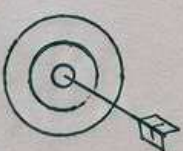
- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- RMSE



Evaluate using various metrics to ensure the model is reliable and effective.

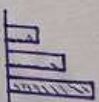
Objective

Build accurate and reliable models that generate insights and drive better decisions.



Key Benefits

★ ★ ★



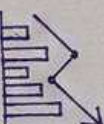
Understand data better



Uncover patterns & trends



Build accurate models



Make data driven decisions